Project 4 — Designing Fair Comparisons

Comparing two groups responsibly.

Project 4: Comparison of two groups

Total points: 40, Phase B graded deliverables: 35 (one-page report) + 5 (R notebook). Phase A (the ten study-design questions) is an optional milestone.

In this project, you will apply study design, DAGs, simulated data, and R (tidyverse) to a research question that your instructor assigns from the course set of questions.

Project workflow

Phase	What	Points
A, Optional worksheet	The ten study-design short answers (Part 1)	0 (ungraded)
B, Full project	One reproducible R notebook (simulate + analyze) and a one-page study-plan report.	40 (35 report + 5 notebook)

Your question. Use the exact question you were assigned from the instructor's list (see Canvas). If the assigned wording allows more than two groups or is ambiguous, narrow it to a single binary explanatory contrast (two groups) in consultation with your instructor if needed.

Sample questions:

- Are people with older siblings who attended college more likely to attend college than people without older siblings who attended college?
- Does sleep quality relate to success at pickle ball?
- Do students learn more at small-group tables than in rows?
- Does homework policy relate to achievement?
- Do hours playing video games relate to academic performance?

For a sample project, see Project 4, Example: yoga in the second trimester and gestational diabetes (worked one-pager) and R notebook (open in Colab). The sample is a single two-group comparison (yoga yes/no) of the response variable, maternal diabetes or not.

Phase A (optional worksheet): Ten study-design questions

This phase is an ungraded worksheet (0 points). Treat it as preparation for the report. Answer using your assigned research question. Use vocabulary from Module 4 Read 1 (study design, two-group tests/CIs, and confounding).

1. Variables. Identify the **explanatory** and **response** variables in your assigned question. For each, give an operational definition: exactly what you measure, units if numerical, and levels or cutoffs for categories (so another student could record data the same way).
2. Observational/Experimental units. What are the **people** or **items** from which data will be collected? Note: data from each of these items should put on the rows of the dataset with a column for each variable.
3. Means or proportions? Will you summarize the comparison with a **difference in means** (numerical response) or a **difference in proportions** (binary / two-category response)?

4. Hypotheses. State **null and alternative hypotheses** with symbols for the population quantities (e.g. μ_1 and μ_2 for means, or p_1 and p_2 for proportions). Define each symbol in a sentence in the context of the research question (e.g. μ_1 is the mean height of soy bean plants grown with fertilizer type A or p_1 is the proportion of radishes eaten by gophers in the winter).
5. RCT. Describe how you would run an ideal **randomized controlled trial** to answer the question, even if it is not feasible in real life. Be sure to mention random *assignment* of the groups in the explanatory variable along with when and what you would measure as the response variable.
6. RCT: two explanations. If the data from an RCT showed a difference between the two groups, give the two general types of explanations for that finding.
7. Observational study. Describe how an **observational study** could be carried out: who you observe, when, and what you record (note that in this design there is no random assignment of the explanatory groups).
8. Observational: three explanations. If the observational study showed a difference in responses between the two explanatory groups, give the three general types of explanations.
9. Confounding and DAG. Name one plausible **confounder** for the observational comparison. Draw a DAG with three nodes (explanatory, response, confounder) and arrows you could defend. In words, explain how the confounder is related to both the explanatory variable and the response.
10. Statistical test for "by chance" explanation. Using the Methods table, name the appropriate test to address the "by chance" explanation for your variable types. State the sample-size / expected-count and other conditions you must check before trusting the test's p-value.

Phase B, Code notebook (5 points)

Goal. One reproducible notebook that (1) simulates a small, realistic-looking dataset for the research question and (2) analyzes it with appropriate descriptive statistics (numerical summaries and a plot) and inferential methods (test/CI). See the sample R notebook for the expected structure (simulate → describe → check the confounder → test)

The simulated dataset should have:

- One row per observational unit; columns for the binary explanatory variable, the response variable, and, if a RCT is not feasible, the confounding variable you identified.
- Enough data for large-sample rules (e.g. $n \geq 20$ per group for t-tests; 2×2 counts with expected 5 for proportions/chi-square tests.

Calculate descriptive statistics for the simulated dataset using:

- Numerical summaries: `group_by` + `summarize` for means/SDs; `count` / `mutate` for counts and proportions.
- Plots: boxplots/histograms by explanatory group for a numerical response; 100% stacked bar charts by explanatory group for a categorical response.
- If you have a confounder, use `facet_wrap(~ confounder)` in plots and include it in `group_by` for the numerical summaries.

Perform the appropriate inferential method for the simulated dataset to see whether the data is consistent with the "by chance explanation".

- Numerical response: `t.test(response ~ explanatory, data = mydataset, var.equal = FALSE)`, report p-value and 95% CI.
- Binary categorical response: `prop.test` / `chisq.test`
- If you have a confounder, then the test doesn't help very much because we can't distinguish between confounding and causation.

Code notebook rubric (5 points)

A notebook (`.ipynb` / `.qmd` / `.Rmd`) that runs top to bottom and includes:

Points	Criterion
2	Reproducible simulation of a realistic dataset with `set.seed()`: one row per observational unit, columns for the binary explanatory variable, the response, and the confounder (if no RCT), with enough data for large-sample rules.
2	Descriptive analysis by explanatory group: `group_by` + `summarize` (means/SDs) or `count` + `mutate` (counts/proportions), plus an appropriate plot (boxplots/histograms for a numerical response; 100% stacked bars for a categorical response; `facet_wrap(~ confounder)` if used).
1	Correct inferential method run on the simulated data with reported output: `t.test` or `prop.test`/`chisq.test`, showing the p-value and 95% CI.

Phase B, One-page study-plan report (35 points)

Write a one page summary of your recommended study plan for a classmate. Include your recommended analysis performed on simulated data.

- **Introduction**: Restate the research question from your classmate and describe the purpose of this document (which is to give a plan for collecting, analyzing and making an appropriate conclusion based on the data; sample R code is also provided and tested on simulated data).
- **Plan to Collect Data**. Describe the response and explanatory variables, units, and measurement (see your answers to questions 1 and 2 from part 1). Justify your recommendation of either a RCT or an observational study based on the considerations of real-world feasibility and our desire for high quality evidence.
- **Plan to Analyze Data**. In a few sentences describe how to analyze the dataset. Give the names of the plots and numerical summaries you recommend . Give the name of the appropriate inferential method (t-test or proportion z test/chi-square and state the conditions for it's validity. Give the null and alternative hypotheses for the test in symbols, with a sentence describing what each symbol represents (see your answer to questions 3 and 4)
- **Plan to Make an Appropriate Conclusion**. Discuss plausible explanations (causal, confounding, chance) consistent with your recommended study design. When discussing the "by chance" explanation and trying to eliminate it via a test, discuss the two types of errors that can result. That is, assuming that a significance level of 0.05 is used as the cutoff for determining whether a p-value is small, there is a 5% chance of making a Type I error. Say what a Type I Error would be in this context and how to reduce it (reduce significance level. Say what a Type II Error would be in this context and how to reduce it (increase sample size)
- **Simulated Results**: Give the numerical summaries, plots, p-value and CI from your analysis of the simulated data. Make an appropriate conclusion based on these about what this data says about the answer to your classmates question. Give a link to your code notebook (in Colab, go to "share" in the upper right hand corner, set the access to be "with anyone with the link" with view access, copy this link and insert it into your one page summary)

One-page report rubric (35 points)

Points	Criterion
4	Introduction: restates the assigned research question and the purpose of the document (a plan to collect, analyze, and conclude, tested on simulated data).

Points	Criterion
7	Plan to collect: defines the explanatory and response variables with operational definitions/units/levels and the observational units; recommends an RCT or an observational study with a justification (feasibility vs. quality of evidence).
8	Plan to analyze: names the recommended plots and numerical summaries matched to the variable types; names the appropriate inferential method (Welch two-sample t-test, or two-proportion z/chi-square test) and states its conditions; states H_0 and H_a in symbols with a sentence defining each symbol in context.
8	Plan to conclude: discusses the plausible explanations consistent with the chosen design (causal / confounding / chance); defines a Type I and a Type II error in context and how to reduce each (lower the significance level; increase the sample size).
6	Sample results: reports the numerical summaries, plot(s), p-value, and 95% CI from the simulated data, makes an appropriate conclusion, and includes a working Colab share link to the notebook.
2	Format: one side of a standard 8.5×11 in. page, complete sentences.

What to turn in to Canvas

Phase A (optional worksheet, 0 points):

- Answers to the ten study-design questions

Phase B (graded, 40 points):

1. One-page report (file upload): a single PDF that follows the one-page rubric above (35 points).
2. R code notebook (file upload): one notebook (`.ipynb`) that runs top to bottom, including simulation of the data and all analysis code and plots (5 points).